

2) Todos los polinomios de Bernoulli de orden 0 son polinomios mónicos elementales. $B_0^0(x) = x^0$

$$\varphi(0; \tau; x) = \frac{\tau^0 e^{\tau x}}{(e^{\tau} - 1)^0} = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_v^0(x)$$

$$\sum_{r=0}^{\infty} \frac{(\tau x)^r}{r!} = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_v^0(x)$$

$$\sum_{r=0}^{\infty} \frac{\tau^r}{r!} x^r = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_v^0(x)$$

Luego: $x^r = B_r^0(x) \quad r \in \mathbb{N}_0$